

Motivation - Research Gap

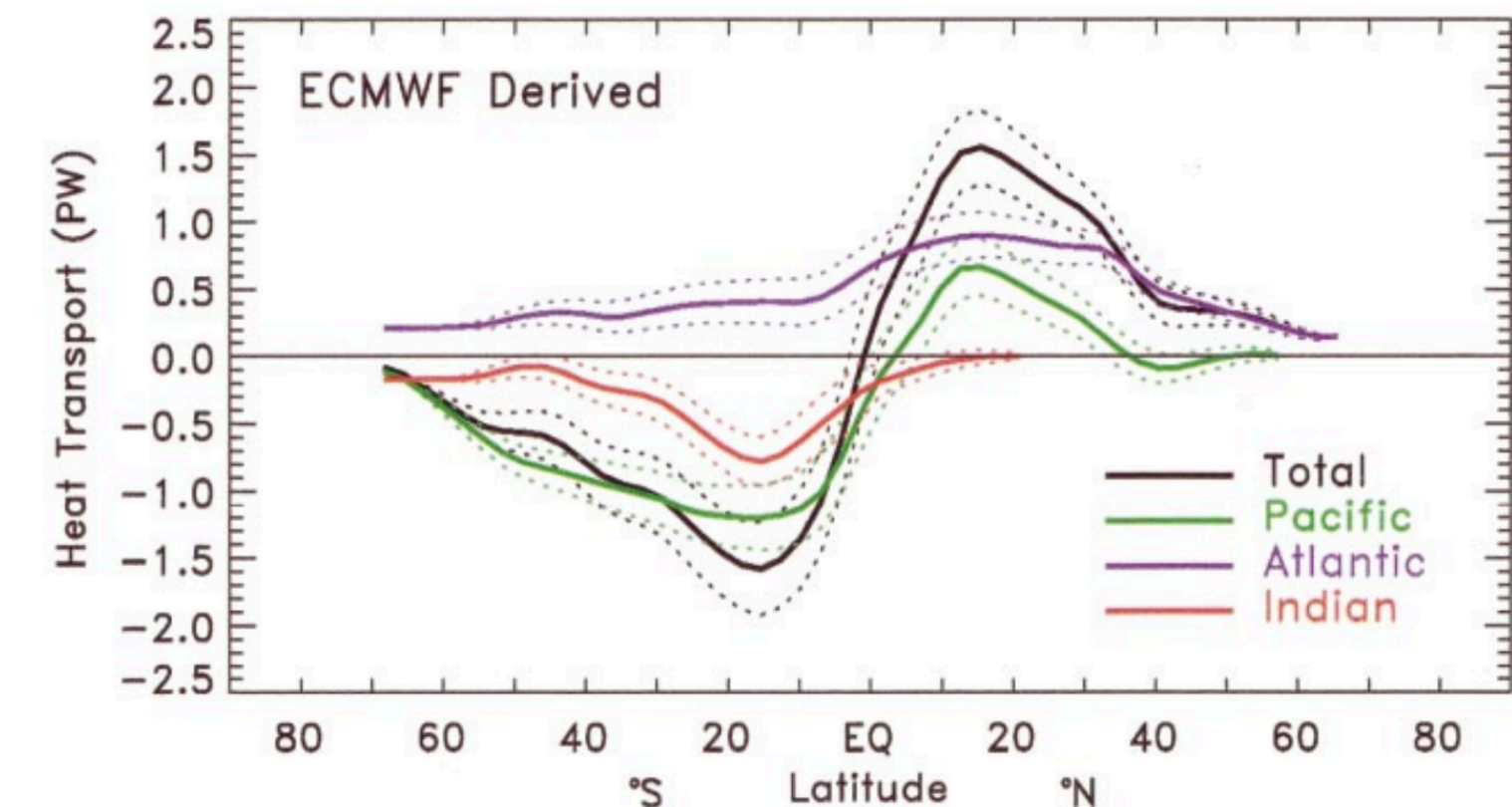
Why do we care about heat transport in the Atlantic Ocean?

- Heat flux into ocean can result either in warming or in circulation / transport changes
 - The heat gets distributed globally by the ocean currents and stored
 - At places where the heat is again released can affect local surface climate
 - For example position of ITCZ depends on MHT over the equator in the atlantic - its displacement can cause heavy rain in some regions while droughts in others
- Variability of MHT on these timescales is still unknown
- Important to distinguish on longer timescales what is inherent variability vs. what is long-term human induced trend

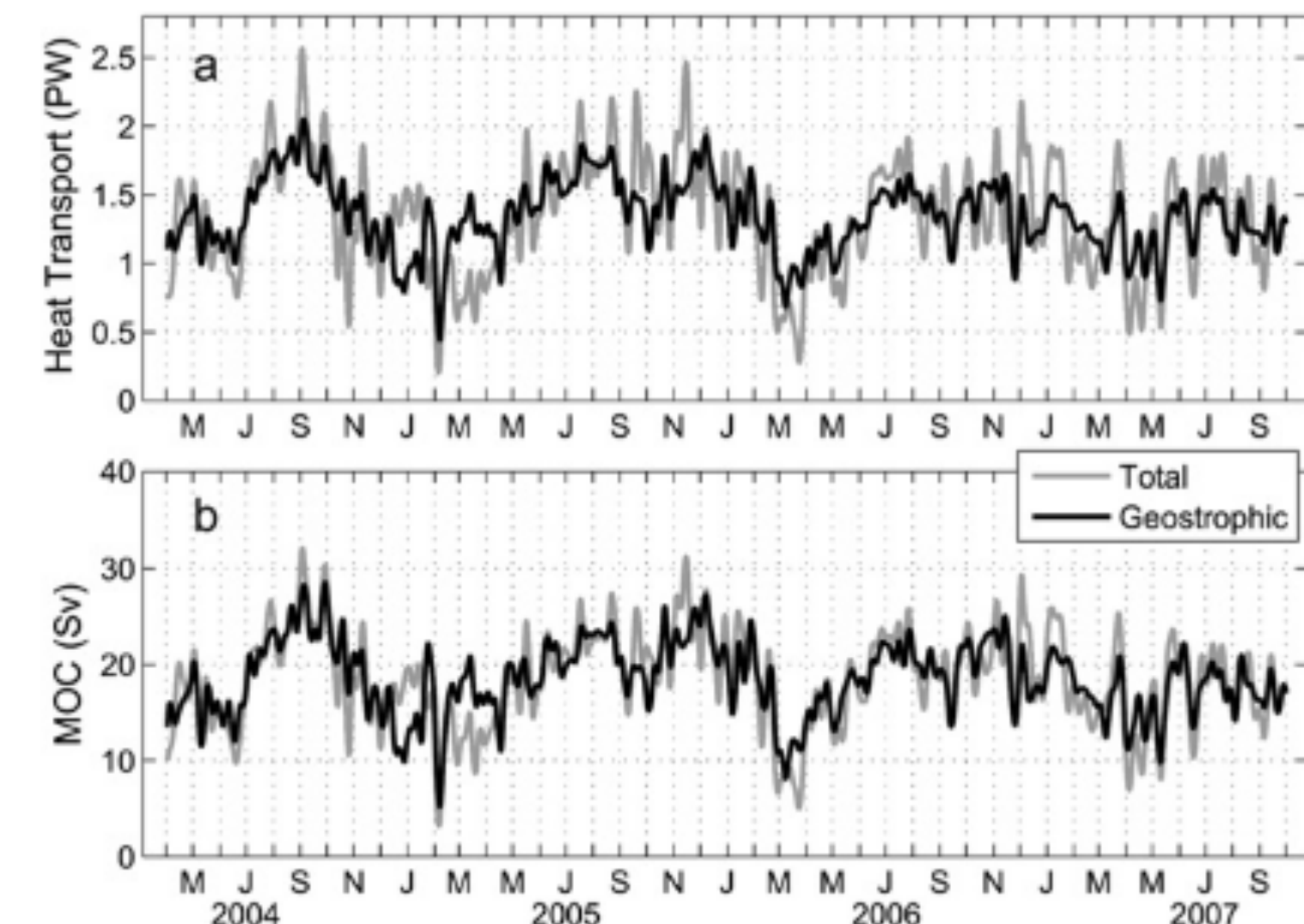
Motivation - Research Gap

Why do we care about heat transport in the Atlantic Ocean?

- Here **AMOC** (Atlantic Meridional Overturning Circulation) and the Atlantic come into play!
- **Atlantic is unique:** heat transport is northward at all latitudes
- Johns et al. 2011 found **nearly linear relationship** between MHT-AMOC: **0.065 PW/Sv** and
- AMOC carries around ~88% of total MHT in atlantic



Trenberth and Caron (2001)



Johns et al. (2011)